آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Pacing Guide for Teachers

CHEMISTRY

GRADE IX

Theory

FOR ACADEMIC YEAR 2025 AND ONWARDS

Number of weeks: 28

Number of periods per week: 5

Key Textbook: *Chemistry for Grade 9,* National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Teacher Developer(s) and their Institution(s):

Mrs. Mairaj Kamran – The Mama Parsi Girls' Secondary School & Aga Khan Higher Secondary School, Karachi

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Topic	Total Periods
1. Fundamentals of Chemistry	11

Sub-Topic	Range of SLOs	Periods (40 mins)
1.1 Chemistry and its Branches	1.1.1 – 1.1.2	3
1.2 History of Chemistry	1.2.1 – 1.2.3	2
1.3 Basic Definitions, Comparisons, Valencies, and Chemical Formulae	1.3.1 – 1.3.2	2
	1.3.3 – 1.3.4	2
	1.3.5 – 1.3.7	2

FOR ACADEMIC YEAR 2025 AND ONWARDS

Suggested Activities and/ or Formative Assessment

Activity 1:

Ask students to make a concept map of branches of chemistry.

Activity 2:

Provide students with a blank diagram or template and ask them to draw and label the structures of atoms, ions, molecules, compounds, and mixtures.

Activity 3:

Ask students to gather information about the history of chemistry and share it in group discussions.

Activity 4:

Link for worksheets and concepts

https://www.liveworksheets.com/w/en/chemistry/1844495

https://www.mychemistryclass.net/Files/Extra%20Resources%20for%20Website/Writing%20Names%20and%20Formulas%20WS%201.pdf

http://www.natureofchemistry.com/theories-and-paradigm-shifts.html

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

Further Resource

For additional resources related to teaching, learning and formative assessments, please refer to: **Learn Smart Classroom by Knowledge Platform:**

https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
2. Stoichiometry	17

Sub-Topic	Range of SLOs	Periods (40 mins)
2.1 Avogadro's Number and Mole	2.1.1 – 2.1.2	2
	2.1.3	2
2.2 Formulae and Percentage Composition	2.2.1 – 2.2.2	2
	2.2.3 – 2.2.4	2
2.3 Chemical Reactions and Calculations	2.3.1 – 2.3.3	2
	2.3.4	2
	2.3.5	2
	2.3.6	3

FOR ACADEMIC YEAR 2025 AND ONWARDS

Suggested Activities and/ or Formative Assessment

Activity 1:

Ask students to make a flow chart to recall the formulae and interconversion of moles, mass, and molecules for the numerical/ unitary method.

Activity 2:

Worksheet for numerical of number of moles, mass, molecules, and Avogadro's number.

Links for worksheets and activities

https://studylib.net/doc/8588062/avogadro-s-number-worksheet

https://www.chemistrylearner.com/worksheets/types-of-chemical-reactions-worksheets

https://learningcentre.vcc.ca/media/vcc-library/content-assets/learning-centre/worksheets/by-coursex2fprogram/mathx2fscience/Chem0861-EmpiricalFormulasAndMolecularFormulas.pdf

https://phet.colorado.edu/en/simulations/balancing-chemical-equations

Learning Resource

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ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
3. Atomic Structure	15

Sub-Topic	Range of SLOs	Periods (40 mins)
3.1 Features of an Atom	3.1.1 – 3.1.2	2
	3.1.3 – 3.1.4	3
3.2 Isotopes	3.2.1 – 3.2.2	2
	3.2.3 – 3.2.4	2
3.3 Models to Understand the Structure of an Atom	3.3.1 – 3.3.3	3
3.4 Shells and Sub-shells	3.4.1	1
3.5 Electronic Configuration	3.5.1	2

FOR ACADEMIC YEAR 2025 AND ONWARDS

Suggested Activities and/ or Formative Assessment

Activity 1:

Link for activity

https://phet.colorado.edu/sims/html/build-an-atom/latest/build-an-atom_en.html

Activity 2:

Atomic structure worksheet

https://www.gpb.org/sites/default/files/2020-04/atomstructurewkst.pdf

Activity 3:

Electronic configuration- YouTube link

https://youtu.be/hSkJzE2Vz_w

Activity 4:

Electronic configuration – worksheet link

https://studylib.net/doc/25800766/electronic-structure-worksheet

Activity 5:

Interactive worksheet on shells and sub-shells

https://www.liveworksheets.com/qz2319995ke

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

Further Resource

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FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
4. Periodic Table and Periodicity	16

Sub-Topic	Range of SLOs	Periods (40 mins)
4.1 Periodic Table	4.1.1 – 4.1.5	3
4.2 Periodic Properties	4.2.1 – 4.2.2	3
	4.2.3	3
4.3 Properties of Elements in a Group	4.3.1 – 4.3.2	2
	4.3.3 – 4.3.6	4
	4.3.7 – 4.3.8	1

FOR ACADEMIC YEAR 2025 AND ONWARDS

Suggested Activities and/ or Formative Assessment

Activity 1:

Family characteristics

Identify each family in the periodic table. Research and make a list of the common characteristics, properties, and trends associated with each family. Create a visual representation of each family by organizing index cards or pieces of paper with the name of the family, its common properties, and examples of elements belonging to that family. A colour code for each family can be used for easier distinction. Arrange the cards in the order they appear in the periodic table.

Activity 2:

Element showcase

Choose a specific family/ group from the periodic table that you find interesting or want to learn more about. Research and select a few representative elements from that family. Create a mini-exhibit or showcase for each element, including information about its atomic number, symbol, atomic mass, electronic configuration, common uses, and any notable properties or characteristics.

Activity 3:

Family scavenger hunt

Create a scavenger hunt activity focused on specific families/ groups of the periodic table. Develop a list of clues or questions related to the properties or characteristics of elements within those families. Hide the clues around a room or outdoor space and invite participants to find and answer the clues using the periodic table.

Activity 4:

Link for worksheets

https://sciencenotes.org/periodic-table-worksheet-2/

https://www.chemistrylearner.com/worksheets/periodic-table-worksheets

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

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FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
5. Structure of Molecules	22

Sub-Topic	Range of SLOs	Periods (40 mins)
5.1 Formation of Chemical Bond	5.1.1 – 5.1.2	2
	5.1.3 – 5.1.4	2
5.2 Ionic Bond	5.2.1 – 5.2.2	2
	5.2.3	1
	5.2.4	1
5.3 Covalent Bond	5.3.1 – 5.3.3	3
	5.3.4 – 5.3.5	3
	5.3.6	1
5.4 Coordinate Covalent Bond	5.4.1 – 5.4.2	2
	5.4.3	1
5.5 Metallic Bond	5.5.1 – 5.5.2	2
5.6 Intermolecular Forces	5.6.1	2

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resource

https://youtu.be/lP_EsVY4CVg

Suggested Activities and/ or Formative Assessment

Activity 1:

Ask students to create concept maps that illustrate the various types of chemical bonds (ionic, covalent, and metallic) and their characteristics.

Activity 2:

Present students with a set of molecular models or diagrams and ask them to identify the type of bond (ionic, covalent, or metallic) present in each model.

Activity 3:

Organise a group activity where students discuss and debate different aspects of chemical bonding. Assign roles to each student, such as an advocate for ionic bonding, covalent bonding, or metallic bonding.

Following is the hands-on activity on chemical bonding to demonstrate ionic bonding using simple materials. Here's a step-by-step guide:

Materials needed:

- Small squares of aluminum foil
- Small squares of plastic wrap
- Salt (table salt or sodium chloride)
- Water
- Small cups or containers
- Stirring rod or toothpick

Instructions:

- Fill two small cups or containers with water.
- Place a small square of aluminum foil in one cup and a small square of plastic wrap in the other.
- Sprinkle a small amount of salt onto each of the squares.
- Observe the behavior of the salt on the aluminum foil and the plastic wrap.
- Stir the contents of each cup gently with a stirring rod or a toothpick.
- Observe the changes in the water and the behavior of the salt particles.
- Explain the observations based on the concepts of ionic bonding.

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Explanation: The aluminum foil represents metal, while the plastic wrap represents non-metal. When salt (sodium chloride) is added to water, it dissociates into positive sodium ions (Na^+) and negative chloride ions (Cl^-). In the cup with aluminum foil, the aluminum atoms lose electrons and form positive ions (Al^{3+}), which are attracted to the negative chloride ions. This attraction results in the formation of a thin layer of white solid on the aluminum foil, representing the formation of an ionic compound (aluminum chloride).

In the cup with plastic wrap, the plastic is non-conductive and does not allow the flow of electrons, preventing the formation of ions. Therefore, no visible reaction occurs on the plastic wrap.

This hands-on activity provides a visual representation of ionic bonding by demonstrating the attraction between positive and negative ions, highlighting the basic principles of chemical bonding.

Activity 4:

Link for worksheet

https://www.wlwv.k12.or.us/cms/lib8/OR01001812/Centricity/Domain/1638/bonding%20practice%20both%20types.pdf

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

Further Resource

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https://akueb.knowledgeplatform.com/login

FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
6. States of Matter	17

Sub-Topic	Range of SLOs	Periods (40 mins)
6.1 Kinetic Molecular Theory and Phase Changes	6.1.1 – 6.1.2	2
	6.1.3 – 6.1.4	2
6.2 Gaseous State	6.2.1 – 6.2.2	2
6.3 Laws Related to Gases	6.3.1 – 6.3.3	3
6.4 Liquid State	6.4.1 – 6.4.2	2
	6.4.3 – 6.4.4	2
6.5 Solid State	6.5.1 – 6.5.2	2
6.6 Types of Solid	6.6.1 – 6.6.2	2

FOR ACADEMIC YEAR 2025 AND ONWARDS

Web Resource

https://phet.colorado.edu/sims/html/states-of-matster-basics/latest/states-of-matter-basics_en.html

Suggested Activities and/ or Formative Assessment

Activity 1:

Use graphic organisers like Venn diagrams, T-charts, or flowcharts to compare the properties of solids, liquids, and gases. Have students fill in the organiser with relevant information and provide feedback on their work.

Activity 2:

A phase diagram can be given to identify and interconvert states of matter.

Activity 3:

Links for activities and worksheets

https://phet.colorado.edu/en/simulations/diffusion

https://phet.colorado.edu/en/simulations/gases-intro

https://phet.colorado.edu/en/simulations/gas-properties

https://phet.colorado.edu/en/simulations/states-of-matter

https://www.wrschool.net/cms/lib/AZ02214740/Centricity/Domain/1625/States%20of%20Matter.pdf

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

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FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
7. Solutions	19

Sub-Topic	Range of SLOs	Periods (40 mins)
7.1 Introduction to Solutions	7.1.1	1
7.2 Types of Solution According to Phases	7.2.1	1
7.3 Types of Solution According to Concentration	7.3.1 – 7.3.2	1
7.4 Comparison of Solution, Suspension and Colloid	7.4.1	1
7.5 Concentration Units and Dilution of Solutions	7.5.1 – 7.5.2	3
	7.5.3 – 7.5.5	3
7.6 Factors Affecting Solubility	7.6.1 – 7.6.2	2
	7.6.3 – 7.6.4	2
7.7 Methods for Separating Mixtures and Purification Techniques	7.7.1 – 7.7.5	5

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Suggested Activities and/ or Formative Assessment

Activity 1:

Present students with real-life scenarios involving solubility issues, such as dissolving a particular substance in each solvent under specific conditions. Ask them to analyse the situation, identify the relevant factors affecting solubility, and propose solutions or explanations.

Activity 2:

A practical activity related to the composition of a solution is “The Dissolving Race” (refer to the given link). This activity allows students to explore the factors that affect the rate at which a solute dissolves in a solvent.

https://www.youtube.com/watch?v=6S3Qjr6Ym7I

Activity 3:

Prepare solutions of different concentrations to find the molarity and present its report including appropriate units, observations, and errors.

Activity 4:

Give students a word bank to sort out solutions, suspensions, and colloids in their respective columns.

Activity 5:

Use Venn diagrams to compare the properties of solutions, suspensions, and colloids. Have students complete it with relevant information and provide feedback on their work.

Activity 6:

Links for activities and worksheets

Chromatography can be performed in school using tissue paper.

https://www.liveworksheets.com/gf2791720nj

https://phet.colorado.edu/en/simulations/concentration

https://www.labxchange.org/library/items/lb:LabXchange:be52b2cf:lx_simulation:1

https://www.chemistrylearner.com/molarity-worksheets/molarity-and-dilution-worksheets

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FOR ACADEMIC YEAR 2025 AND ONWARDS

Topic	Total Periods
8. Electrochemistry	23

Sub-Topic	Range of SLOs	Periods (40 mins)
8.1 Oxidation and Reduction (Redox) Reactions	8.1.1	1
8.2 Oxidation States and Rules for Assigning Oxidation States	8.2.1 – 8.2.3	3
8.3 Oxidising and Reducing Agents	8.3.1 – 8.3.2	2
8.4 Electrochemical Cells	8.4.1 – 8.4.2	2
	8.4.3 – 8.4.5	3
	8.4.6 – 8.4.7	2
	8.4.8 – 8.4.9	2
	8.4.10	1
8.5 Electrochemical Industries	8.5.1 – 8.5.2	4
8.6 Corrosion and its Prevention	8.6.1 – 8.6.3	3

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Web Resources

https://www.labxchange.org/library/items/lb:LabXchange:DEL5DErJ:html:1

https://www.labxchange.org/library/items/lb:LabXchange:8b3f4dc3:video:1

Suggested Activities and/ or Formative Assessment

Activity 1:

Provide students with unbalanced redox equations and ask them to balance them to understand the process of assigning the oxidation numbers and balancing half-cell reactions.

Activity 2:

Provide students with real-life scenarios where oxidation or reduction reactions occur, such as corrosion of metals, combustion reactions, or biological redox reactions. Ask them to analyse the situation, identify the oxidising and reducing agents, and explain the overall process.

Activity 3:

Ask students to compare galvanic cells and electrolytic cells, including similarities, differences in operation, and the purpose of each type of cell.

Activity 4:

Students can be assessed with a diagram of an electrochemical cell where they could label the anode, cathode, direction of electron flow, direction of ion flow, and the overall cell reaction.

Activity 5:

Set up corrosion experiments to demonstrate how metals react with oxygen in the presence of moisture.

Link for worksheets

https://www.liveworksheets.com/az2007154ce

https://quizlet.com/au/368946834/corrosion-flash-cards/

Learning Resource

AKU-EB Teaching and Learning Resource Guide:

https://examinationboard.aku.edu/about-us/SyllabiList/Resource%20Guide%202025%20SSC%20Chemistry.pdf

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Note: This teacher-led pacing guide has been developed by educators from AKU-EB affiliated schools to support fellow teachers teaching the AKU-EB syllabi by

- ensuring the smooth transition of a school's academic year.
- ensuring curricular continuity in schools.
- predicting the time and pace of syllabi implementation.

This document also contains **suggested activities and/ or formative assessments** that may enhance the learning experience. Please note that these activities are meant to serve as suggestions. As educators, you have the flexibility and autonomy to adapt and modify them to best suit the needs of your students and the dynamics of your classroom.

You are advised to use an ad-blocker while accessing websites and web resources. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine.

FOR ACADEMIC YEAR 2025 AND ONWARDS